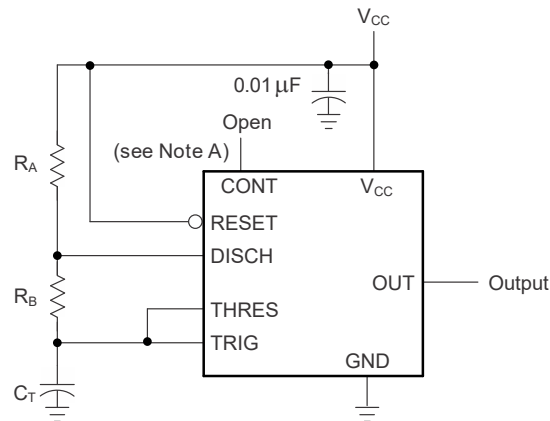




Decouple CONT voltage to ground with a capacitor to improve operation. Reevaluate for individual applications.

Figure 6-3. Circuit for Astable Operation

$$t_H \cong 0.693 \times (R_A + R_B) \times C_T \quad (1)$$

$$t_L \cong 0.693 \times R_B \times C_T \quad (2)$$

Other useful relationships for period, frequency, and driver-referred and waveform-referred duty cycle are shown as follows:

$$T = t_H + t_L \cong 0.693 \times (R_A + 2R_B) \times C_T \quad (3)$$

$$f = \frac{1}{T} \cong \frac{1.44}{(R_A + 2R_B) \times C_T} \quad (4)$$

$$\text{Output driver duty cycle} = \frac{t_L}{T} \cong \frac{R_B}{R_A + 2R_B} \quad (5)$$

$$\text{Output waveform duty cycle} = \frac{t_H}{T} \cong 1 - \frac{R_B}{R_A + 2R_B} = \frac{R_A + R_B}{R_A + 2R_B} \quad (6)$$

6.4 Device Functional Modes

Table 6-1 shows the device truth table. For a valid reset voltage condition, use an external pullup resistor to V_{CC} (if using the RESET functionality), or short the RESET pin directly to V_{CC} (if the RESET functionality is not used).

Table 6-1. Function Table

RESET VOLTAGE ⁽¹⁾	TRIGGER VOLTAGE ⁽¹⁾	THRESHOLD VOLTAGE ⁽¹⁾	OUTPUT	DISCHARGE SWITCH
LOW	Irrelevant	Irrelevant	Low	On
> MAX	< $1/3 \times V_{CC}$	Irrelevant ⁽²⁾	High	Off
> MAX	> $1/3 \times V_{CC}$	> $2/3 \times V_{CC}$	Low	On
> MAX	> $1/3 \times V_{CC}$	< $2/3 \times V_{CC}$	As previously established	

(1) Voltage levels shown are nominal.

(2) CONT pin open or $2/3 \times V_{CC}$.